SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Short Answers

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – Short Answers
Weightage:	30 % of CIA	**Total Marks:**	100 Marks
CO1 Statement:	*Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
Student Name:	CH. Ishwarya	**Reg. No.:**	192324145
Section / Batch:	AI&DS	**Date:**	9/11/26

Code of Conduct

I CH. Ishwarya (192324145) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: CH. Ishwarya

Instructions

- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

Faculty In-charge	Course Coordinator	Head of Department
Dr. TERRANCE		
Signature:	Signature:	Signature:
Date: 09/11/26	Date:	Date:

94/100

9/10/26

Assessment Tool - 2
Mobile computing

Name: CH. Ishwarya
Reg No: 192324145

1) spectrum auction = 45MHz = 45000MHz
full-duplex = 25x2 = 50
full-duplex voice channel = $\frac{45000}{50}$
= 900 channel

2) 560/7 = 80 Channels

3) Total voice channels
100x8 = 800

4) 10 log 10 (P0/4)
= 10log 10 (20)
= 13.01dB

5) ~~Total~~
C/I = -60 - (-92)
= 32 dB

6) Total channel = 42000KHz
Channels = $\frac{42000}{300}$ = 140
= 140 channels = 140/7 = 20

7) power = 64w
64/16 = 4w

8) $P \propto d^2$. Dt distance double
scale = $2^2 = 4$
= 4

9) 4x15 = 60 channels

10) v = 100/3.6 = 27.78m/s
fD = v/l = (27.78x900x10)/(3x10)
= 83.3Hz

11) 5/8 = 0.625 ms

10) $24{,}000 / 200 = 120$ carriers

13) $8W = 8000\ mW$

$10\log_{10}(8000) = 39.03\ dBm$

Received $= 39.03 - 33 = 6.03\ dM$

14) $FSPL = 32.44 + 20\log_{10}(1900) + 20\log_{10}(2.5)$

$= 105.98dB \sim 106\ dB$

15) $6 \times 250 = 15000$ cells

16) $3\ min = 180\ s$

Active cells $= 4000 \times 180$

$= 720000$

17) $150 \times 25\ KHz$

$= 3750\ KHz = 3.75\ MHz$

18) 30 bands create 29 gaps blw them

$29 \times 40\ KHz = 1160\ KHz = 1.16\ MHz$

19) Diameter $= 2 \times 15 = 3\ km$

Handoff/hour $= 90/3 = 30$

20) $8 \times 40 \times 250 = 80{,}000$

21) $R = 0.41\ cm$

Hexagonal area $= (3\sqrt{3}/2)R^2 = 0.4157\ km^2$

cells $= 400/0.4157 = 962.25$

22) $6000/30 = 20$ channels

23) $1200 \times 4 \times (2.5/60) = 200$

24) $40{,}000/400 = 100$

25) $1200 \times 600 = 720000$

26) $200/4 = 50$

Offered traffic = 40,000 × 3 = 120,000

Capacity = 100,000. Blocked = 20,000

Blocked fraction = 20,000 / 120,000 × 100
= 16.67%.

28) 600,000 × 0.16 = 96,000

29) 12000 - 9000 = 3000

30) 48,000 / 200 = 240

Cell = 240/4 = 60 channels

31) 30,000 / 200 = 150 channels

32) 2500 × 0.99 = 2,475

33) 12W = 12,000 mw

10log10 (12,000) = 40.79dBM

Received = 40.79 - 45 = -4.21dBM

34) 7 × 900 = 6,300

35) 4.8/8 = 0.6

36) v = 150 / 3.6 = 41.67 m/s

fD = v f / c = (41.67 × 1.8 × 10) / (3 × 10)
= 250 Hz

37) offered load = 70,000 × 4 = 280,000

Blocked = 280000 - 250,000 = 30,000
= 30,000 / 280,000 × 100 = 10.71%.

38) 9 × 600 = 5,400

39) 60,000 × 0.004 = 240

40) path loss = 43 - (-87) = 130dB

41) $150 \times 2.5 = 375$

42) Total channels $= 48{,}000 / 200 = 240$

per cell $= 240 / 6 = 40$ channels

43) $15{,}000 / 6{,}000 = 2.5$

44) $9 / 3 = 3$

45) $-85 - (-115) = 30\,dB$

46) 250×150 kbps $= 37{,}500$ kbps

$= 37.5$ Mbps

47) $1500 \times 4 \times (2.5/60) = 250$

48) 250×12 kHz $= 3000$ kHz

$= 3$ MHz

49) 1500×600 kbps $= 90{,}000$

$= 900$ Mbps

50) Area $= \pi R^2 = \pi(4^2) = 50.27\,km^2$

Towers $= 350 / 50.27 = 6.96$ so

$= 7$ towers.